

GO-DRiVeS Mobile App for Passenger-Vehicle Real-Time Interaction with In-Campus Golf Car

Problem / Gap

- Many applications for ride-hailing exists (Uber, Kareem, etc), but none like Go-Drives.
- AV tracking and monitoring was previously researched, but an app was never implemented.
- No research specifically on a mobile app for in-campus AV ride requests & tracking.

Objectives

- **Main objective:** to develop a mobile application that facilitates real-time interaction between the passenger and the campus golf car.
- **Sub-Objectives:** to develop the following features
 - Ride-Requesting
 - Real-Time Tracking
 - Sensor data monitoring
 - Camera Streams
 - Status Updates

Propositions / Planned Contribution

- Evaluate the existing prototype to identify strengths and areas for improvement.
- Research mobile app development frameworks (e.g., Flutter, React Native) for optimal performance.
- Investigate sensors & communication protocols (e.g., MQTT, WebSockets, 4G/5G, Zigbee, LoRa) for real-time AV data transmission.
- Integrate real-time tracking & ride-request functionality using GPS & cloud-based services (Firebase).
- Implement sensor data monitoring to display AV diagnostics (speed, battery, location, etc.).
- Develop camera streaming functionality for real-time monitoring.